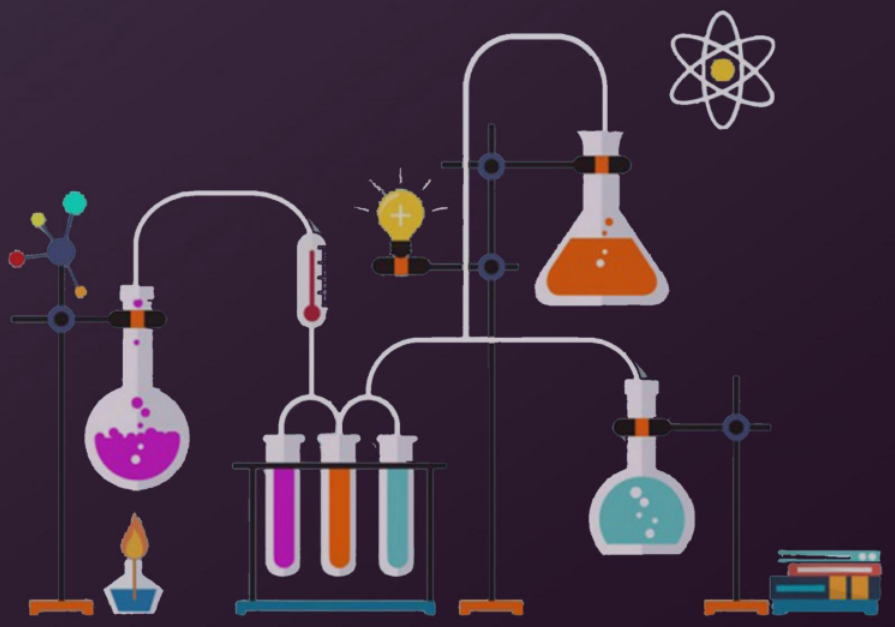
CHEMISTRY

Environmental Chemistry



SHEET

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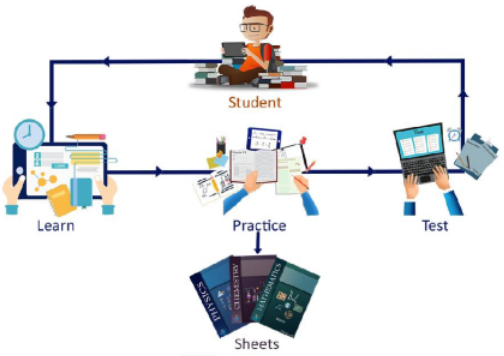
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-2
1	Types of Pollution	1-4	1-3
2	Oxides of Sulphur, Nitrogen and Hydrocarbons	5	4
3	Oxides of Carbon, Global Warming and Greenhouse Effect	6-7	
4	Acid Rain and Taj Mahal	8-10	5
5	Particulate Pollutants and Photochemical Smog	11-12	
6	Stratospheric Pollution and Ozone Hole	13-14	
7	Water Pollution	15-17	6,7
8	Soil Pollution	18-20	
9	Green Chemistry	21-22	
10	ACC	23	8

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Type of Pollution

Q1. The term biosphere is used for the zone of the earth where life exists
(A) On the lithospere
(B) In the hydrosphere
(C) In the lithosphere and hydrosphere
(D) In the lithosphere, hydrosphere and atmosphere

Q2. When biosphere turns into human dominated environment it is called
(A) Noosphere
(B) Troposphere
(C) Mesosphere
(D) Man sphere

Q3. What is the correct sequence of atmospheric layers starting from earth
(A) Stratosphere troposphere, mesosphere, thermosphere
(B) Troposphere, startosphere, mesosphere, thermosphere
(C) Mesosphere, troposphere, stratosphere, thermosphere
(D) Thermosphere, mesophere, stratosphere, troposphere

Q4. Petroleum resource is -
(A) Renewable
(B) Non renewable
(C) Synthetic & biodegradable
(D) Infinite & unconventional

Oxides of Sulphur, Nitrogen and Hydrocarbons

Q5. Main air pollutant among the following is
(A) CO (B) CO_2
(C) N_2 (D) Sulphur

Oxides of Carbon, Global Warming and Greenhouse Effect

Q6. Which will not cause any atmospheric pollution
(A) Hydrogen
(B) Sulphur dioxide
(C) Carbon dioxide
(D) Carbon monoxide

Q7. An increase in CO_2 concentration in the atmosphere will result in
(A) Adverse effects of natural vegetation
(B) Global warming
(C) Temperature decrease in global atmosphere
(D) Genetic disoders in plants and animals

Acid Rain and Taj Mahal

Q8. Recent reports of acid rains industrial cities are due to the effect of atmospheric pollution by :
(A) Excessive release of NO_2 and SO_2 by burning of fossil fuels.
(B) Exessive release of CO_2 by burning of fuel like wood and charcoal, cutting of forests and increased animal population.
(C) Excessive release of NH_3 by industrial plants and coal gas.
(D) Excessive release of CO in atmosphere by incomplete combustion of cock, charcoal and other carbonaceous fuels in pancity of oxygen.

Q9. Acid rain occurs due to atmospheric pollution of

(A) SO_2 (B) NH_3

(C) CO_2 (D) N_2O

Q10. Acid rain is due to increase in atmospheric concentration of

(A) Ozone and dust

(B) CO_2 and CO

(C) SO_3 and CO

(D) SO_2 and NO_2

Particulate Pollutants and Photochemical Smog

Q11. Which one of the following statements regarding photochemical smog is not correct?

(A) Photochemical smog is formed through photochemical reaction involving solar energy

(B) Photochemical smog does not cause irritation in eyes and throat.

(C) Carbon monoxide does not play any role in photochemical smog formation

(D) Photochemical smog is an oxidising agent in character

Q12. Which one of the following is **not** a common component of Photochemical Smog?

(A) Ozone

(B) Acrolein

(C) Peroxyacetyl nitrate

(D) Chlorofluorocarbons

Stratospheric Pollution and Ozone Hole

Q13. Ozone layer of upper atmosphere is being destroyed by

(A) Sulphurdioxide

(B) Carbondioxide

(C) Chlorofluorocarbon

(D) Smog

Q14. Identify the incorrect statement from the following:-

(A) Oxides of nitrogen in the atmosphere can cause the depletion of ozone layer.

(B) Ozone absorbs the intense ultraviolet radiation of the sun.

(C) Depletion of ozone layer is because of its chemical reactions with chlorofluoro alkanes.

(D) Ozone absorbs infrared radiation.

Water Pollution

Q15. Which of the following is the main factor of water pollution

(A) Smoke

(B) Industrial waste

(C) Detergent

(D) Ammonia

Q16. Which is more important for water pollution

(A) Sound

(B) SO_2

(C) Salts of arsenic

(D) Sewage

Q17. If water pollution continues at its present rate, it will eventually

(A) Stop water cycle

(B) Prevent precipitation

(C) Make oxygen molecules unavailable to water plants.

(D) Make nitrate molecules unavailable to water plants.

Soil Pollution

Q18. Pollution can be controlled by

(A) Sewage treatment

(B) Checking atomic blasts

(C) Manufacturing electrically operated vehicles

(D) All the above

Q19. If a lake is contaminated with DDT, its highest concentration would be found in
(A) Primary consumer
(B) Secondary consumer
(C) Tertiary consumer
(D) None of these

Q20. Soil erosion is greater when
(A) No rain occurs
(B) Winds do not blow
(C) The rainfall is evenly distributed
(D) The Rainfall is received in heavy down pour

Green Chemistry

Q21. Maximum green house gas released by which country
(A) India (B) France
(C) U.S.A (D) Britain

Q22. Which of the following is not a green house gas:-
(A) Hydrogen
(B) Carbon dioxide
(C) Methane
(D) Nitrous oxide or N_2O

ACC

Q23. Which is not a renewable source
(A) Forest
(B) Coal
(C) Water
(D) Forest organism

Exercise - 2 *Concept Builder*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Biosphere is -
(A) In which individual interact to each other
(B) By which life originated
(C) The name of a bird
(D) Organic compound by which life diminishes

Q2. The living organisms on or around the earth consitute
(A) Biome
(B) Biosphere
(C) Community
(D) Biocoenosis

Q3. Air pollutants that produce photochemical smog –
(A) Ozone, chlorine and sulphur dioxide
(B) Oxygen, chlorine and nitric acid
(C) Nitrous oxide, PAN and acrolein
(D) CO_2, CO and SO_2

Q4. Which of the following atmospheric pollutants is not produced by the exhaust of motor vehicle in Delhi ?
(A) SO_2
(B) Hydrocarbon gases
(C) Fly ash
(D) CO

Q5. Taj Mahal is threatened by pollution from –
(A) Carbon dioxide
(B) Oxygen
(C) Sulphur dioxide
(D) Chlorine

Q6. The major source of BOD in the river Ganga is -

(A) Leaf litter (B) Fishes

(C) Human waste (D) Aquatic plants

Q7. What is B.O.D.

(A) The amount of O_2 utilised by organisms in water

(B) The amount of O_2 utilized by micro organisms for decomposition

(C) The total amount of O_2 present in water

(D) All of the above

Q8. PAN (Peroxy Acyl Nitrate) is an important constituent of photochemical smog, is a–

(A) Quantitative pollutant

(B) Primary pollutant

(C) Natural pollutant

(D) Secondary pollutant

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	A	B	B	A	A	B	A	A	D
11	12	13	14	15	16	17	18	19	20
B	D	C	D	B	C	C	D	C	D
21	22	23							
C	A	B							

Ex-2							
1	2	3	4	5	6	7	8
A	B	C	A	C	C	C	D